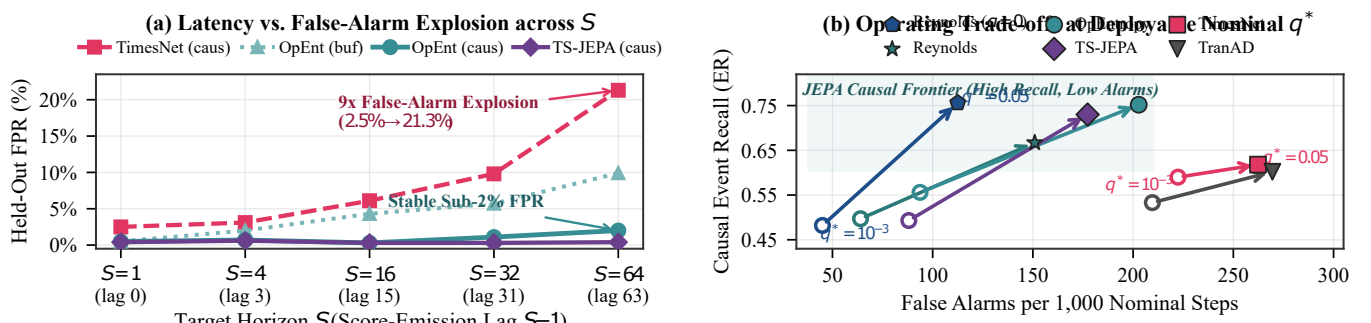

(a) Latency vs. False-Alarm Explosion across S
TimesNet (caus)
OpEnt (buf)
OpEnt (caus)
TS-JEPA (caus)
9x False-Alarm Explosion
(2.5%→21.3%)
Stable Sub-2% FPR
Held-Out FPR (%)
20%
15%
10%
5%
0%
$S=1$
(lag 0)
$S=4$
(lag 3)
$S=16$
(lag 15)
$S=32$
(lag 31)
$S=64$
(lag 63)
Target Horizon S(Score-Emission Lag $S-1$)
(b) Operating Trade-off at Deployable Nominal q^*
RiemTS (caus)
OpEnt (caus)
Reynolds
TS-JEPA
TranAD
JEPA Causal Frontier (High Recall, Low Alarms)
$q^* = 0.05$
$q^* = 10^{-3}$
$q^* = 0.05$
$q^* = 10^{-3}$
Causal Event Recall (ER)
0.75
0.65
0.55
0.45
50
100
150
200
250
300
False Alarms per 1,000 Nominal Steps